Justin S. Derenthal

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EDUCATION

Farmingdale State College | Long Island, NY **May 2026**
***B.S. Computer Science* | GPA: 3.81/4.0**

- **Relevant Coursework:** Data Structures & Algorithms I-II, Operating Systems, Computer Networks, Computer Architecture, Data Management (SQL), Software Engineering, Principles of Programming Languages, Applied Probability & Statistics
- **Honors:** President's List; Dean's List; 3x 1st Place – Cybersecurity Capture the Flag (CTF)

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn), Java, SQL, JavaScript, TypeScript, Kotlin

Frameworks & Tools: FastAPI, Node.js, React, Azure SQL, Firebase, MongoDB, Microsoft Power Platform, MediaPipe, Git, Linux

EXPERIENCES

DataAnnotation - Remote **August 2023 – 2026**
Software Developer – AI Consultant

- Evaluated and audited thousands of large language model outputs for **accuracy, formatting, logical reasoning, and factual correctness** across diverse subject domains
- Designed structured prompts and **JSON**-based response specifications for advanced mathematics, **Python, Java,** and computer vision-based tasks
- Classified model behaviors and curated datasets to improve evaluation reliability and training data quality
- Contributed to proprietary model evaluation workflows to assess reasoning consistency and improve response reliability

Farmingdale State College – Department of Computer Science **January 2024 – Present**
Teaching Assistant - Data Structures & Algorithms I, Programming: Python, Programming: Java

- Delivered instruction to **200+** students across multiple semesters in **Java** and **Python**, covering recursion, arrays, lists, stacks, heaps, and algorithmic time/space complexity
- Developed structured lesson plans and led instructional sessions reinforcing implementation strategy and debugging methodology
- Collaborated with faculty to refine and optimize programming assignments to improve conceptual clarity and learning outcomes
- Held weekly office hours to guide students through complex algorithmic problem-solving

Farmingdale State College – Office of Distance Learning **October 2025 – Present**
Distance Learning Help Desk Technician

- Engineered automated ticket notification and reminder workflows using **Microsoft Power Automate,** integrating **Microsoft Forms** with **ManageEngine ServiceDesk Plus**
- Designed dynamic auto-response templates using variable-based logic to improve response consistency and reduce repetitive manual effort
- Reduced ticket completion time and standardized communication across campus-wide helpdesk operations
- Currently developing **Power BI** dashboards to analyze service trends and identify workflow optimization opportunities

Video Game Trading Post – Long Island, NY **August 2018 – September 2024**
General Manager

- Directed multi-location retail operations, overseeing staffing coordination, logistics, and financial reconciliation
- Leveraged multi-store performance analytics to optimize product allocation and improve inventory turnover
- Managed vendor relationships and purchasing strategy to align supply with regional demand trends
- Spearheaded physical setup and operational launch of new store locations from ground-up execution

PROJECTS

OpenHand – Real Time ASL Recognition System – *Python, FastAPI, Scikit-learn, MediaPipe, Kotlin, React, Firebase*

- Architected a real-time ASL recognition pipeline using **MediaPipe** (21 landmarks) and **RandomForestClassifier** trained on tens of thousands of curated samples
- Achieved **~95%** real-time classification accuracy with live inference via **FastAPI** and **WebSockets** across **Android** and web clients
- Implemented **Firebase**-based authentication, progress tracking, and leaderboard systems for persistent user learning

Circuit Simulator – Event-Driven Electrical Modeling System – *Java/JavaFX, MySQL, Azure Blob Storage*

- Engineered **20,000+ line** object-oriented simulation modeling voltage/current flow with event-driven state propagation and cloud-based persistence
- Designed relational database schema and **JSON** serialization structure to support normalized data persistence

SUNY Payroll Data Pipeline & Visualization (In progress) – *Python, Pandas, NumPy, Power BI*

- Building a structured data ingestion and normalization pipeline in **Python (Pandas, NumPy)** with schema standardization to aggregate SUNY payroll datasets, delivering interactive **Power BI** dashboards for compensation analysis
- Implementing data validation and consistency checks to ensure cross-institution compensation reporting accuracy

TikTok Share Prediction – Machine Learning Study – *Python, Pandas, Scikit-learn*

- Developed **Logistic Regression** model on **19,000+ records** to predict post virality, achieving **87.92% classification accuracy** using K-fold cross-validation and feature standardization